
Cambridge Assessment
International Education

Question	Answer	Marks	Partial Marks
1(a)	$\frac{5}{6}$ oe	**1**	
1(b)	[0].08 oe	**1**	
2(a)	Table and pictogram correct 8 12 5 7 Apple  Orange 	**3**	**B1** for 12 and 7 correct **B1** for Apple row correct **B1** for Orange row correct
2(b)	Banana	**1**	
3	(4, 2) (3, 4)	**2**	**B1** for each or for two correct points plotted
4(a)	21	**1**	
4(b)	–5.5	**1**	
5(a)	18.2 to 19	**2**	**B1** for 9.1 to 9.5 [cm] seen or **M1** for *their* distance in cm × 2
5(b)	071 to 075	**1**	
6	15	**2**	**M1** for $250 \times \frac{1.5}{100} \times 4$ oe
7	8.15	**2**	**B1** for answer figs 815 or for 0.85 seen or 900 seen
8	145	**3**	**M1** for $(5 - 2) \times 180$ oe **M1** for *their* $540 - (110 + 125 + 65 + 95)$
9(a)	14 17	**1**	
9(b)	$3n + 2$ oe final answer	**2**	**B1** for $3n + k$ oe seen or for $jn + 2$ oe seen $j \neq 0$
9(c)	12	**3**	**M1** for substitution of 20 into *their* $(3n + 2)$ or for 62 seen or for 38 seen **M1** for *their* $(3k + 2) = 100 -$ *their* 62

Question	Answer	Marks	Partial Marks
10	$\frac{40}{71}$	**3**	**B2** for 15 : 40 : 16 oe seen or for 40 : 71 or **M1** for attempt to equate number of blue in the two ratios or for 71 seen or for $\frac{r}{g}=\frac{15}{16}$ soi
11(a)	Acceptable bisector of $\angle PSR$ with correct arcs	**2**	**B1** for acceptable bisector with no/incorrect arcs or for short bisector with arcs
11(b)	Correct region shaded.	**1**	**FT** *their* **(a)** dependent on attempt at angle bisector
12(a)	[0].00204 or 2.04×10^{-3} cao	**1**	
12(b)	4, 60 and 0.4 seen **and** final answer 300	**2**	**B1** for two of 4, 60 and 0.4 seen
13(a)	144	**2**	**B1** for $(3 \times 4)^2$ or $3^2 \times 4^2$ or better
13(b)	Irrational number in range $4.5 \leqslant n \leqslant 5.5$	**1**	
14(a)	2×10^{-4}, 0.002, 2×10^{-2}, 2000	**1**	
14(b)	[a =] 2 [b =] 8	**2**	**B1** for each or **SC1** for $a = 20$ and $b = 9$
15	18	**2**	**B1** for $k = 2$ if $y = k(x - 1)^2$ used or **M1** for $\frac{32}{(5-1)^2}=\frac{y}{(-2-1)^2}$ or **M1FT** for y = *their* $k \times (-2 - 1)^2$
16(a)	Tangent drawn at $x = -1$	**B1**	
	–3 to –2	**B1**	Dep on close attempt at tangent at $x = -1$
16(b)	–3.9 to –3.8 0 3.8 to 3.9	**3**	**B1** for each If 0 scored, **M1** for line $y = 2$ drawn at least from (–1, 2) to (1, 2) If 0 scored, **SC1** for answers (–3.9 to –3.8, 2) and (0, 2) and (3.8 to 3.9, 2)
17(a)	Square Rhombus	**1**	

Question	Answer	Marks	Partial Marks
17(b)	Two correct pairs of angles from $\angle AXC = \angle BXD$, $\angle CAX = \angle DBX$ and $\angle ACX = \angle BDX$ with correct reasons **and** $AX = BX$ with correct reason **and** ASA stated e.g. $AX = BX$ given $\angle AXC = \angle BXD$ [vertically] opposite $\angle CAX = \angle DBX$ alternate [angles] [Congruent] ASA	**3**	**B2** for two pairs of equal sides/angles with correct reasons or **B1** for one pair of equal sides/angles with correct reason or for two appropriate pairs with no or incorrect reasons
18	$\frac{x+7}{3}$ oe final answer	**2**	**B1** for $x = 3y - 7$ or $y + 7 = 3x$ or $\frac{y}{3} = x - \frac{7}{3}$
19(a)(i)	{a, e, f, g, h, i, j}	**1**	
19(a)(ii)	6	**1**	
19(b)	$A \cap C \cap B'$ oe	**1**	
20(a)	35	**1**	
20(b)	100	**2**	**B1** for $\angle OAB = 55$ or $\angle OBA = 55$
21(a)	$(4x - 3)(x + 2)$ final answer	**2**	**M1** for brackets giving two out of three terms correct in a quadratic expression or for $4x(x + 2) - 3(x + 2)$ seen or for $x(4x - 3) + 2(4x - 3)$ seen
21(b)	$\frac{x^3}{4}$ final answer	**2**	**B1** for answer $\frac{x^3}{k}$ or $\frac{x^k}{4}$ or for $\left(\frac{x^6}{16}\right)^{\frac{1}{2}}$ seen or for $\left(\frac{4}{x^3}\right)^{-1}$ seen
22(a)	$\frac{7}{9}$ oe	**1**	
22(b)	$\frac{1}{9}$ or $\frac{8}{72}$ oe nfww	**3**	**M2** for $\frac{3}{9} \times \frac{2}{8} + \frac{2}{9} \times \frac{1}{8}$ oe or **M1** for $\frac{3}{9} \times \frac{2}{8}$ or $\frac{2}{9} \times \frac{1}{8}$ seen If 0 scored, **SC1** for answer $\frac{13}{81}$ or $\frac{8}{81}$
23	7π final answer	**2**	**M1** for $\frac{360 - 80}{360} \times \pi \times 3^2$ oe If 0 scored, **SC1** for answer 2π

Question	Answer	Marks	Partial Marks
24(a)	–2	**3**	**M1** for $2 - 5x = 3(3x + 10)$ or better **M1** for $2 - 30 = 9x + 5x$ or better
24(b)	$\frac{x+13}{(x-2)(2x+1)}$ or $\frac{x+13}{2x^2-3x-2}$ final answer	**3**	**B1** for $3(2x + 1) - 5(x - 2)$ oe isw **B1** for denominator $(x - 2)(2x + 1)$ oe isw
25(a)	$\frac{5}{2}\mathbf{c}$ oe final answer	**1**	
25(b)	$4\mathbf{a} + \frac{3}{2}\mathbf{c}$ or $\frac{8\mathbf{a}+3\mathbf{c}}{2}$ final answer	**2**	**B1** for final answer $4\mathbf{a} + k\mathbf{c}$ oe or $k\mathbf{a} + 1.5\mathbf{c}$ oe , $k \neq 0$ or $\overrightarrow{AP} = 3\mathbf{a}$ soi or **M1** for a correct route along the lines of the diagram using the given vertex letters
25(c)	1 : 10 oe	**1**	